

Stability-Preserving Sparse H_2 Control with Mandatory Structural Constraints via Proximal Gradient Methods

Wenliang Sheng, Chao Yang, Kemi Ding

Abstract—This paper investigates sparse H_2 state-feedback design for continuous-time linear systems under partial structural constraints, where a prescribed subset of feedback links is required to remain active. The resulting problem is nonconvex and nonsmooth due to the coexistence of closed-loop stability, ℓ_1 -regularized sparsity promotion, and mandatory nonzero structural requirements. To address this difficulty, we introduce a lower-bound reformulation of the mandatory nonzero constraints and develop a stability-preserving proximal gradient method for the reformulated problem. We further establish convergence properties of the algorithm and show that every accumulation point of the generated sequence is a stationary point of the reformulated problem. Numerical experiments on both small-scale and large-scale systems demonstrate that the proposed method effectively captures the sparsity–performance trade-off while preserving mandatory feedback links, and achieves more stable convergence and higher computational efficiency than an ADMM-based alternative.

Index Terms— H_2 performance, proximal gradient methods, sparse control, structural constraints.

I. INTRODUCTION

In large-scale and networked control systems, sparse feedback design has attracted significant attention because it offers an effective trade-off between closed-loop performance and controller complexity [1], [2], [3]. By eliminating redundant feedback links, sparse controllers can reduce communication overhead and implementation burden, especially in distributed and communication-constrained settings. However, in many practical applications, such as power systems, multi-agent networks, and industrial process control, the controller structure is not entirely free [4], [5], [6]. Due to physical interconnections, communication requirements, and safety considerations, certain feedback links are indispensable and must be preserved. These considerations naturally motivate the study of sparse controller design under partial structural constraints.

Existing works can be broadly categorized into two directions. The first direction focuses on sparsity-promoting

controller design, where sparsity is induced by incorporating regularization terms into the control objective so that redundant feedback links can be automatically eliminated [7], [8], [9], [10], [11], [12], [13], [14], [15]. Lin *et al.* developed sparse and block-sparse synthesis based on the alternating direction method of multipliers (ADMM), using ℓ_1 , weighted ℓ_1 , cardinality, and nonconvex sum-of-logs penalties [7], [8]. Matni *et al.* used architecture-inducing atomic regularization in a unified model-matching framework [9]. Babazadeh *et al.* further improved sparsity promotion using ℓ_1 and reweighted ℓ_1 mechanisms [11], while Bahavarnia *et al.* used quasi-norm regularization to provide a tighter approximation of cardinality and stronger sparsification than standard ℓ_1 penalties [12]. Despite their effectiveness, these approaches typically generate sparse patterns implicitly and do not explicitly guarantee preservation of prescribed mandatory links. The second direction considers controller design under prescribed structural constraints, where the admissible controller architecture is specified a priori according to physical interconnections, communication topologies, or decentralized information patterns [6], [16], [17], [18]. In this line of research, the prescribed structure may itself be sparse, but the sparsity pattern is treated as a prior architectural constraint rather than being induced automatically during optimization. Wu proposed an optimization-based approach for static output-feedback sliding mode control under zero–nonzero structural constraints, where sparsity is achieved by enforcing a given structural pattern while maintaining stability and performance requirements [19]. Yang *et al.* studied structural linear quadratic regulators under prescribed sparse patterns, where both state-feedback and static output-feedback synthesis are formulated as constrained optimization problems subject to structural constraints on the controller gain matrix, and proposed a tractable first-order log-barrier search method with convergence guarantees to stationary solutions [20]. Although such methods can yield sparse controllers under prescribed architectures, they typically assume that the admissible sparsity pattern is specified a priori, thereby restricting the structural flexibility available during optimization and limiting the ability to exploit the remaining structural freedom for further closed-loop performance improvement.

Therefore, existing studies either promote sparsity through regularization without explicitly preserving prescribed mandatory links, or impose a pre-specified structural architecture without exploiting the remaining structural freedom to improve

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the sparsity–performance trade-off. This reveals a gap in the existing literature on sparse controller design under partial structural constraints, where certain prescribed feedback links must remain active while the remaining entries should be optimized to achieve a desirable sparsity–performance trade-off.

Motivated by the above observations, this paper investigates sparse H_2 state-feedback design under partial structural constraints, where a prescribed subset of feedback gains must remain active. This problem is inherently nonconvex and nonsmooth, due to the interplay among closed-loop stability, sparsity promotion, and mandatory nonzero structural requirements. To handle these difficulties, we propose a stability-preserving proximal gradient framework based on a tractable lower-bound reformulation of the mandatory nonzero constraints. The proposed method integrates ℓ_1 -regularized sparsity promotion with explicit preservation of prescribed feedback links, while a backtracking line search is used to guarantee closed-loop stability along the iteration process. We show that the reformulated problem is feasible for sufficiently small lower-bound parameters and establish that every accumulation point of the generated iterates is a stationary point of the constrained sparse H_2 optimization problem. Numerical examples further demonstrate the effectiveness and computational advantages of the proposed method.

The main contributions of this paper are summarized as follows. First, we formulate a sparse H_2 state-feedback design problem under partial structural constraints, which explicitly captures the practical requirement that certain indispensable feedback links must remain nonzero. Second, we propose a stability-preserving proximal gradient method together with a numerically tractable lower-bound reformulation of the mandatory nonzero constraints, so that sparsity promotion and structural preservation can be handled within a unified optimization framework. Third, we establish feasibility and convergence properties of the proposed method, and numerical results on both small-scale and large-scale systems show that it achieves competitive sparsity–performance trade-offs with improved computational efficiency and more stable convergence behavior compared with an ADMM-based method.

The remainder of this paper is organized as follows. Section II formulates the problem. Section III presents the proposed stability-preserving proximal gradient method and its analysis. Section IV provides numerical results. Section V concludes the paper.

II. PROBLEM SETUP

A. Notation

For a matrix A , $A^\top$ denotes its transpose, and $\text{tr}(A)$ denotes its trace. $\mathbb{R}^{m \times n}$ denotes the set of all $m \times n$ real matrices. For a matrix $K \in \mathbb{R}^{m \times n}$, K_{ij} denotes its (i, j) -th entry. The Frobenius norm and the entrywise ℓ_1 -norm are denoted by $\|\cdot\|_F$ and $\|\cdot\|_1$, respectively. For a symmetric matrix X , $X \succ 0$ and $X \succeq 0$ mean that X is positive definite and positive semidefinite, respectively. A square matrix is said to be Hurwitz if all its eigenvalues have negative real parts. For a square matrix M , $\lambda_i(M)$ denotes its i -th eigenvalue, and $\Re(\lambda_i(M))$ denotes the real part of $\lambda_i(M)$. For

a scalar $x \in \mathbb{R}$, $\text{sign}(x)$ denotes its sign. For an index set Ω , Ω^c denotes its complement. The notation $\partial f(\cdot)$ denotes the limiting subdifferential of a function f , and $\text{prox}_{ng}(\cdot)$ denotes the proximal mapping associated with a proper lower semicontinuous function g .

B. System Model

We consider the following continuous-time linear time-invariant (LTI) system:

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) + Ew(t), \\ z(t) &= Cx(t) + Du(t), \end{aligned} \quad (1)$$

where $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the control input, $w(t) \in \mathbb{R}^{p_w}$ is an exogenous disturbance, and $z(t) \in \mathbb{R}^{p_z}$ is the performance output. The matrices A , B , C , D , and E are constant matrices of compatible dimensions. Throughout the paper, we assume that the pair (A, B) is stabilizable and the pair (A, C) is detectable.

We consider static state-feedback controllers of the form

$$u(t) = Kx(t), \quad K \in \mathbb{R}^{m \times n}, \quad (2)$$

where K is the feedback gain matrix to be designed.

Substituting (2) into (1) yields the closed-loop system

$$\begin{aligned} \dot{x}(t) &= A_{\text{cl}}x(t) + Ew(t), \\ z(t) &= C_{\text{cl}}x(t), \end{aligned} \quad (3)$$

where

$$A_{\text{cl}} := A + BK, \quad C_{\text{cl}} := C + DK.$$

If A_{cl} is Hurwitz, then the squared H_2 norm of the closed-loop transfer function $T_{zw}(s)$ from w to z is given by

$$J_{H_2}(K) := \|T_{zw}(s)\|_{H_2}^2 = \text{tr}(C_{\text{cl}}XC_{\text{cl}}^\top) = \text{tr}(E^\top YE), \quad (4)$$

where $X \succ 0$ and $Y \succ 0$ are the unique solutions to the Lyapunov equations

$$A_{\text{cl}}X + XA_{\text{cl}}^\top + EE^\top = 0, \quad (5)$$

and

$$A_{\text{cl}}^\top Y + YA_{\text{cl}} + C_{\text{cl}}^\top C_{\text{cl}} = 0. \quad (6)$$

Accordingly, the standard H_2 state-feedback design problem is to determine a gain matrix K that minimizes $J_{H_2}(K)$ while ensuring that $A + BK$ is Hurwitz.

C. Structural Constraints

In many large-scale and networked control systems, the feedback gain matrix K is not entirely free to be designed. Due to physical interconnections, communication requirements, and safety considerations, certain feedback links must remain active and cannot be removed. For example, in distributed sensor–actuator networks, some sensor–actuator connections may be indispensable for safety or reliability reasons. At the same time, not all feasible feedback links are equally important. Even when additional feedback channels are implementable, a dense feedback architecture may significantly increase communication, computation, and implementation costs.

These considerations motivate a sparse controller design paradigm under partial structural constraints, where indispensable feedback links are explicitly preserved while sparsity is promoted over the remaining entries of the feedback gain matrix. To characterize such mandatory connections, let

$$\Omega_M \subseteq \{1, \dots, m\} \times \{1, \dots, n\}$$

denote the index set of mandatory feedback entries. Then, for every $(i, j) \in \Omega_M$, the corresponding feedback gain entry K_{ij} is required to remain nonzero.

Accordingly, the controller design problem considered in this paper seeks a sparse feedback gain matrix K such that the closed-loop H_2 performance is optimized, the closed-loop system remains stable, and all mandatory feedback links indexed by Ω_M are preserved.

D. Problem Formulation

Based on the system model and the above structural requirements, we formulate the following sparse H_2 state-feedback design problem under partial structural constraints:

$$\begin{aligned} \min_K \quad & J_{H_2}(K) + \lambda \|K\|_1 \\ \text{s.t.} \quad & A + BK \text{ is Hurwitz,} \\ & K_{ij} \neq 0, \quad \forall (i, j) \in \Omega_M, \end{aligned} \quad (\text{P1})$$

where $\lambda > 0$ is a regularization parameter that balances closed-loop performance and sparsity, and the entrywise ℓ_1 -norm is adopted as a standard sparsity-promoting regularizer.

Problem (P1) is challenging for several reasons. First, the objective function $J_{H_2}(K)$ depends on the feedback gain K implicitly through the closed-loop Lyapunov equations (5)–(6), and is therefore generally nonconvex. Second, the stability requirement that $A + BK$ be Hurwitz defines a nonconvex feasible set in the controller space. Third, the mandatory nonzero constraints $K_{ij} \neq 0$ for $(i, j) \in \Omega_M$ are themselves nonconvex. As a result, Problem (P1) is a nonconvex and nonsmooth optimization problem, and cannot be directly handled by standard convex optimization methods.

Remark 1. When $\lambda = 0$, Problem (P1) reduces to an H_2 state-feedback design problem under mandatory structural constraints only, where sparsity is not explicitly promoted. This special case serves as a natural baseline for evaluating the effect of the ℓ_1 regularization term in the subsequent numerical studies.

III. STABILITY-PRESERVING PROXIMAL GRADIENT METHOD

A. Structural feasibility of mandatory nonzero links

Before developing the optimization algorithm, we first examine the feasibility of the mandatory nonzero structural constraints. In particular, we show that enforcing prescribed feedback entries to remain nonzero does not destroy stabilizability, and that such constraints can be satisfied by arbitrarily small perturbations of any stabilizing feedback gain. This observation provides the basis for the numerical reformulation developed in the next subsection.

To this end, define the stabilizing controller set

$$\mathcal{H} := \{K \in \mathbb{R}^{m \times n} : A + BK \text{ is Hurwitz}\}.$$

We begin with the following basic property.

Lemma 1. *The set $\mathcal{H}$ is open.*

Proof: Let $\alpha(M) := \max_i \Re(\lambda_i(M))$ denote the spectral abscissa of a square matrix M , where $\lambda_i(M)$ denotes the i -th eigenvalue of M . Since eigenvalues depend continuously on the matrix entries, the spectral abscissa $\alpha(\cdot)$ is continuous. Define

$$g(K) := \alpha(A + BK).$$

Because the mapping $K \mapsto A + BK$ is affine, $g(\cdot)$ is continuous in K . Moreover, $A + BK$ is Hurwitz if and only if $g(K) < 0$. Therefore,

$$\mathcal{H} = \{K \in \mathbb{R}^{m \times n} : g(K) < 0\} = g^{-1}((-\infty, 0)).$$

Since $(-\infty, 0)$ is an open set and g is continuous, $\mathcal{H}$ is open. $\square$

The openness of $\mathcal{H}$ implies that any stabilizing controller admits a neighborhood consisting entirely of stabilizing controllers. We next show that this property allows the mandatory nonzero constraints to be enforced through arbitrarily small perturbations.

Let

$$\mathcal{N}(\Omega_M) := \{K \in \mathbb{R}^{m \times n} : K_{ij} \neq 0, \forall (i, j) \in \Omega_M\}$$

denote the set of feedback gains satisfying the mandatory nonzero requirements.

Lemma 2. *Suppose that (A, B) is stabilizable. Then*

$$\mathcal{H} \cap \mathcal{N}(\Omega_M) \neq \emptyset.$$

Moreover, for any $K_0 \in \mathcal{H}$ and any $\varepsilon > 0$, there exists $K^ \in \mathcal{H} \cap \mathcal{N}(\Omega_M)$ such that*

$$\|K^* - K_0\|_F < \varepsilon.$$

Proof: Since (A, B) is stabilizable, there exists at least one stabilizing feedback gain, and hence $\mathcal{H} \neq \emptyset$. Fix any $K_0 \in \mathcal{H}$. By Lemma 1, since $\mathcal{H}$ is open, there exists $\delta > 0$ such that

$$\|K - K_0\|_F < \delta \implies K \in \mathcal{H}.$$

We now construct a perturbation $\Delta \in \mathbb{R}^{m \times n}$ that enforces the mandatory nonzero entries while keeping the perturbation arbitrarily small. For each $(i, j) \in \Omega_M$, define

$$\Delta_{ij} := \begin{cases} 0, & \text{if } (K_0)_{ij} \neq 0, \\ \eta_{ij}, & \text{if } (K_0)_{ij} = 0, \end{cases}$$

where each $\eta_{ij} \neq 0$ will be chosen sufficiently small. For all $(i, j) \notin \Omega_M$, set $\Delta_{ij} = 0$. Let

$$K^* := K_0 + \Delta.$$

By construction, we have

$$K_{ij}^* \neq 0, \quad \forall (i, j) \in \Omega_M,$$

and thus $K^* \in \mathcal{N}(\Omega_M)$.

Since only finitely many entries of Δ are nonzero, we can choose the values η_{ij} sufficiently small such that

$$\|\Delta\|_F < \min\{\delta, \varepsilon\}.$$

It follows that

$$\|K^* - K_0\|_F = \|\Delta\|_F < \delta,$$

which implies $K^* \in \mathcal{H}$. At the same time,

$$\|K^* - K_0\|_F < \varepsilon.$$

Therefore, $K^* \in \mathcal{H} \cap \mathcal{N}(\Omega_M)$, and the proof is complete. $\square$

Lemma 2 shows that the mandatory nonzero structural constraints do not invalidate stabilizability. More importantly, such constraints can be satisfied by arbitrarily small perturbations of any stabilizing feedback gain. This property provides the theoretical foundation for the lower-bound reformulation introduced next.

B. Numerical reformulation with lower-bound constraints

Although the mandatory nonzero constraints in Problem (P1) are conceptually natural, they are nonconvex and not directly amenable to numerical optimization. Motivated by Lemma 2, we next introduce a numerically tractable reformulation by replacing the nonzero constraints with lower-bound constraints on the prescribed feedback entries.

Specifically, for a sufficiently small constant $\kappa > 0$, we consider the following reformulated optimization problem:

$$\begin{aligned} \min_K \quad & J_{H_2}(K) + \lambda \|K\|_1 \\ \text{s.t.} \quad & A + BK \text{ is Hurwitz,} \\ & |K_{ij}| \geq \kappa, \quad \forall (i, j) \in \Omega_M. \end{aligned} \quad (\text{P2})$$

The lower-bound constraints in (P2) explicitly enforce the mandatory entries to stay away from zero, and hence provide a numerically tractable surrogate for the original nonzero requirements in Problem (P1). The following proposition shows that this reformulation preserves feasibility when κ is sufficiently small.

Proposition 1. *For any sufficiently small $\kappa > 0$, Problem (P2) admits at least one stabilizing feasible solution. Moreover, every feasible solution of Problem (P2) is also feasible for Problem (P1).*

Proof: By Lemma 2, for any stabilizing feedback gain $K_0 \in \mathcal{H}$ and any $\varepsilon > 0$, there exists a matrix $K^* \in \mathcal{H} \cap \mathcal{N}(\Omega_M)$ such that

$$\|K^* - K_0\|_F < \varepsilon.$$

Since $K^* \in \mathcal{N}(\Omega_M)$, all entries K_{ij}^* , $(i, j) \in \Omega_M$, are nonzero. Because Ω_M is finite, there exists a positive constant

$$\kappa^* := \min_{(i,j) \in \Omega_M} |K_{ij}^*| > 0.$$

Hence, for any $\kappa \in (0, \kappa^*]$, the matrix K^* satisfies

$$|K_{ij}^*| \geq \kappa, \quad \forall (i, j) \in \Omega_M.$$

Together with $K^* \in \mathcal{H}$, this shows that K^* is feasible for Problem (P2). Therefore, Problem (P2) admits at least one stabilizing feasible solution for all sufficiently small $\kappa > 0$.

Next, let K be any feasible solution of Problem (P2). Then $A + BK$ is Hurwitz and

$$|K_{ij}| \geq \kappa > 0, \quad \forall (i, j) \in \Omega_M.$$

Therefore,

$$K_{ij} \neq 0, \quad \forall (i, j) \in \Omega_M,$$

which means that K also satisfies the constraints of Problem (P1). Hence every feasible solution of Problem (P2) is feasible for Problem (P1). $\square$

Proposition 1 shows that the lower-bound reformulation (P2) is consistent with the original problem in the sense that it preserves stabilizing feasibility for sufficiently small κ and approaches the original mandatory nonzero constraints as $\kappa \rightarrow 0$. In the sequel, we develop a stability-preserving proximal gradient method for solving the reformulated problem (P2).

C. A Stability-Preserving Proximal Gradient Method

We now develop a stability-preserving proximal gradient method for solving the reformulated Problem (P2). Although Problem (P2) replaces the mandatory nonzero constraints in Problem (P1) by lower-bound constraints, it remains a nonconvex and nonsmooth optimization problem due to the coexistence of the closed-loop stability requirement and the ℓ_1 regularization term. Accordingly, we focus on computing locally optimal stabilizing solutions.

To facilitate algorithm design, we rewrite Problem (P2) in a composite form. Define

$$F(K) := J_{H_2}(K) + g(K), \quad (7)$$

where

$$g(K) := \lambda \|K\|_1 + I_C(K), \quad (8)$$

and $I_C(K)$ is the indicator function of the lower-bound constraint set

$$\mathcal{C} := \{K \in \mathbb{R}^{m \times n} : |K_{ij}| \geq \kappa, \forall (i, j) \in \Omega_M\},$$

namely,

$$I_C(K) := \begin{cases} 0, & K \in \mathcal{C}, \\ +\infty, & K \notin \mathcal{C}. \end{cases} \quad (9)$$

With this notation, Problem (P2) can be equivalently written as

$$\min_{K \in \mathcal{H}} F(K), \quad (\text{P2-C})$$

where $\mathcal{H} = \{K \in \mathbb{R}^{m \times n} : A + BK \text{ is Hurwitz}\}$ denotes the stabilizing controller set.

The function $J_{H_2}(K)$ is smooth over $\mathcal{H}$, and its gradient is given as follows.

Lemma 3. *For each $K \in \mathcal{H}$, the H_2 cost $J_{H_2}(K)$ is continuously differentiable on $\mathcal{H}$, and its gradient is*

$$\nabla J_{H_2}(K) = 2(B^\top Y + D^\top C_{cl})X, \quad (10)$$

where $C_{cl} = C + DK$, and $X \succ 0$, $Y \succ 0$ are the unique solutions to the Lyapunov equations (5)–(6).

Proof: The result follows from standard sensitivity analysis of the H_2 performance index with respect to the feedback gain matrix. Since $K \in \mathcal{H}$, the closed-loop matrix $A + BK$ is Hurwitz, and the Lyapunov equations (5)–(6) admit unique positive definite solutions X and Y . Differentiating the H_2 cost with respect to K yields (10). $\square$

The objective function $F(K)$ consists of a smooth but generally nonconvex term $J_{H_2}(K)$ and a nonsmooth, nonconvex term $g(K)$. Nevertheless, $g(K)$ is proper and lower semicontinuous, and its proximal mapping is well defined and computationally tractable due to its separable structure. Based on the composite structure in (7)–(8), we adopt a proximal gradient (PG) framework to solve Problem (P2), following recent advances in nonconvex and nonsmooth composite optimization [21]. Given a stabilizing iterate $K^t \in \mathcal{H}$, one proximal gradient step is defined as

$$K^{t+1} \in \text{prox}_{\eta_t g}(K^t - \eta_t \nabla J_{H_2}(K^t)), \quad (11)$$

where $\eta_t > 0$ is the step size. The gradient $\nabla J_{H_2}(K)$ is well defined for all $K \in \mathcal{H}$ and can be computed via the solutions of the Lyapunov equations, as characterized in Lemma 3. Let

$$V := K^t - \eta_t \nabla J_{H_2}(K^t). \quad (12)$$

Then the update in (11) is obtained by solving

$$K^{t+1} \in \arg \min_{K \in \mathbb{R}^{m \times n}} \left\{ \frac{1}{2\eta_t} \|K - V\|_F^2 + \lambda \|K\|_1 + I_{\mathcal{C}}(K) \right\}. \quad (13)$$

Since both the squared Frobenius norm and the entrywise ℓ_1 norm are separable across the entries of K , and the constraint set $\mathcal{C}$ only imposes entrywise lower-bound conditions on mandatory entries, the proximal subproblem (13) decouples elementwise. Consequently, the update for each entry (i, j) admits the following closed-form expression:

$$K_{ij}^{t+1} = \begin{cases} \text{sign}(V_{ij}) \max(|V_{ij}| - \eta_t \lambda, 0), & (i, j) \notin \Omega_M, \\ \text{sign}(V_{ij}) \max(|V_{ij}| - \eta_t \lambda, \kappa), & (i, j) \in \Omega_M. \end{cases} \quad (14)$$

For non-mandatory entries, (14) reduces to the standard soft-thresholding operator. For mandatory entries, the lower bound κ is explicitly enforced, thereby preserving the prescribed nonzero feedback links while still promoting sparsity over the remaining entries. A detailed derivation of this proximal mapping is provided in Appendix I.

A key difficulty in applying (11) lies in the stability constraint $K^{t+1} \in \mathcal{H}$. Although the proximal step is well defined algebraically, the updated matrix obtained from (14) may fail to preserve closed-loop stability if the step size is chosen too large. To address this issue, we incorporate a stability-preserving backtracking line search.

Specifically, starting from an initial trial step size η_t , we compute the proximal update (14) and check whether the resulting matrix satisfies $A + BK^{t+1}$ is Hurwitz. If this condition is violated, the step size is reduced and the proximal update is recomputed until a stabilizing iterate is obtained. Since the stabilizing set $\mathcal{H}$ is open by Lemma 1, for every $K^t \in \mathcal{H}$ there exists a sufficiently small step size $\eta_t > 0$ such that the resulting update remains in $\mathcal{H}$. Therefore, the

backtracking procedure is guaranteed to terminate after finitely many reductions.

The above construction yields a proximal gradient method that preserves the mandatory lower-bound constraints by design and maintains closed-loop stability through backtracking. In the next subsection, we establish convergence properties of the proposed algorithm and summarize its implementation.

D. Convergence Analysis and Algorithm Summary

We next establish a convergence result for the proposed stability-preserving proximal gradient method. Since Problem (P2-C) is nonconvex and nonsmooth, our goal is to show that every accumulation point of the generated iterate sequence is a stationary point of the reformulated problem.

To this end, suppose that the backtracking line search accepts a step size $\eta_t > 0$ such that the updated iterate K^{t+1} satisfies the sufficient decrease condition

$$F(K^{t+1}) \leq F(K^t) - c \|K^{t+1} - K^t\|_F^2 \quad (15)$$

for some constant $c > 0$ independent of t . We further assume that the sequence $\{K^t\}$ generated by the algorithm is bounded.

Proposition 2. *Let $\{K^t\} \subseteq \mathcal{H}$ be the sequence generated by the proposed stability-preserving proximal gradient method for Problem (P2-C). Suppose that the accepted step sizes satisfy the sufficient decrease condition (15), that there exists $\eta > 0$ such that $\eta_t \geq \eta$ for all t , and that the sequence $\{K^t\}$ is bounded. Then every accumulation point of $\{K^t\}$ is a stationary point of Problem (P2-C).*

Proof: By (15), for all $t \geq 0$,

$$F(K^{t+1}) \leq F(K^t) - c \|K^{t+1} - K^t\|_F^2.$$

Summing the above inequality from $t = 0$ to T yields

$$\sum_{t=0}^T \|K^{t+1} - K^t\|_F^2 \leq \frac{1}{c} (F(K^0) - F(K^{T+1})).$$

Since $F(\cdot)$ is bounded below on the feasible set, it follows that

$$\sum_{t=0}^{\infty} \|K^{t+1} - K^t\|_F^2 < \infty,$$

and hence

$$\|K^{t+1} - K^t\|_F \rightarrow 0. \quad (16)$$

Because $\{K^t\}$ is bounded, there exists at least one accumulation point $\bar{K}$ and a subsequence $\{K^{t_j}\}$ such that $K^{t_j} \rightarrow \bar{K}$.

Next, by the proximal optimality condition associated with (11), there exists $\xi^{t+1} \in \partial g(K^{t+1})$ such that

$$0 = \xi^{t+1} + \nabla J_{H_2}(K^t) + \frac{1}{\eta_t} (K^{t+1} - K^t). \quad (17)$$

Define

$$\zeta^{t+1} := \xi^{t+1} + \nabla J_{H_2}(K^{t+1}) \in \partial F(K^{t+1}).$$

Combining this definition with (17) gives

$$\zeta^{t+1} = \nabla J_{H_2}(K^{t+1}) - \nabla J_{H_2}(K^t) - \frac{1}{\eta_t} (K^{t+1} - K^t). \quad (18)$$

Since all iterates remain in the stabilizing set $\mathcal{H}$ and $\{K^t\}$ is bounded, its closure is a compact subset of $\mathcal{H}$. Because $J_{H_2}(K)$ is continuously differentiable on $\mathcal{H}$, its gradient is Lipschitz continuous on this compact set. Therefore, there exists a constant $L > 0$ such that

$$\|\nabla J_{H_2}(K^{t+1}) - \nabla J_{H_2}(K^t)\|_F \leq L\|K^{t+1} - K^t\|_F.$$

Using this estimate together with (18) and $\eta_t \geq \underline{\eta}$, we obtain

$$\|\zeta^{t+1}\|_F \leq \left(L + \frac{1}{\underline{\eta}}\right)\|K^{t+1} - K^t\|_F.$$

By (16), it follows that

$$\|\zeta^{t+1}\|_F \rightarrow 0.$$

Finally, along the convergent subsequence $\{K^{t_j}\}$, we have

$$K^{t_j} \rightarrow \bar{K}, \quad \zeta^{t_j} \in \partial F(K^{t_j}), \quad \|\zeta^{t_j}\|_F \rightarrow 0.$$

By the closedness of the limiting subdifferential graph, we conclude that

$$0 \in \partial F(\bar{K}),$$

which means that $\bar{K}$ is a stationary point of Problem (P2-C). $\square$

The above result shows that, under standard sufficient decrease and boundedness assumptions, the proposed stability-preserving proximal gradient method converges subsequentially to stationary solutions of the reformulated sparse H_2 design problem. We summarize the complete implementation procedure in Algorithm 1.

Remark 2. The dominant computational cost of each outer iteration lies in solving the two Lyapunov equations (5)–(6) for evaluating the H_2 gradient. The entrywise proximal update in (14) has negligible cost compared with the Lyapunov solves. Hence, the overall per-iteration complexity is mainly determined by the cost of Lyapunov equation solving.

Remark 3. The regularization parameter λ controls the trade-off between closed-loop performance and sparsity, while the lower-bound parameter κ enforces the preservation of mandatory feedback links. In practice, κ should be chosen sufficiently small so as to approximate the original nonzero constraints without overly restricting the feasible set.

IV. NUMERICAL EXPERIMENTS

This section presents numerical experiments to evaluate the effectiveness and scalability of the proposed stability-preserving proximal gradient method. We consider both a small-scale buffer network and a large-scale mass-spring system. For both experiments, we investigate the sparsity-performance trade-off induced by the regularization parameter and visualize the evolution of the feedback sparsity pattern under mandatory structural constraints. In the large-scale setting, we further construct Pareto fronts and compare the proposed PG method with an ADMM-based approach in terms of solution quality, convergence behavior, and computational efficiency.

Algorithm 1 Stability-Preserving Proximal Gradient Method

Require: System matrices A, B, C, D, E ; sparsity weight $\lambda > 0$; mandatory set Ω_M ; lower bound $\kappa > 0$; initial stabilizing gain $K^0 \in \mathcal{H}$; initial step size $\eta_0 > 0$; backtracking factor $\beta \in (0, 1)$; Armijo constant $c > 0$; tolerance $\varepsilon > 0$; maximum iterations $T_{\max}$.

- 1: **for** $t = 0$ **to** $T_{\max} - 1$ **do**
- 2: Solve the Lyapunov equations (5)–(6) at K^t to obtain $X(K^t)$ and $Y(K^t)$;
- 3: Compute $\nabla J_{H_2}(K^t)$ by Lemma 3.
- 4: Set $\eta \leftarrow \eta_0$;
- 5: accept $\leftarrow$ **false**
- 6: **while** not accept **do**
- 7: $\tilde{V} \leftarrow K^t - \eta \nabla J_{H_2}(K^t)$
- 8: **for** each (i, j) **do**
- 9: **if** $(i, j) \notin \Omega_M$ **then**
- 10: $K_{ij}^{t+1} \leftarrow \text{sign}(\tilde{V}_{ij}) \max(|\tilde{V}_{ij}| - \eta\lambda, 0)$
- 11: **else**
- 12: $K_{ij}^{t+1} \leftarrow \text{sign}(\tilde{V}_{ij}) \max(|\tilde{V}_{ij}| - \eta\lambda, \kappa)$
- 13: **end if**
- 14: **end for**
- 15: **if** $A + BK^{t+1}$ is Hurwitz **and** $F(K^{t+1}) \leq F(K^t) - c\|K^{t+1} - K^t\|_F^2$ **then**
- 16: accept $\leftarrow$ **true**
- 17: **else**
- 18: $\eta \leftarrow \beta\eta$
- 19: **end if**
- 20: **end while**
- 21: **if** $\|K^{t+1} - K^t\|_F \leq \varepsilon$ **then**
- 22: **return** K^{t+1}
- 23: **end if**
- 24: **end for**
- 25: **return** K^{t+1}

A. Small-scale buffer network

We first evaluate the proposed stability-preserving sparse controller design method on a small-scale buffer network system described by

$$\dot{x} = Ax + Bu + Ew, \quad z = Cx + Du, \quad (19)$$

where the system matrices are given by

$$A = \text{diag}(-1, -2), \quad B = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \end{bmatrix},$$

$$C = \begin{bmatrix} I_2 \\ 0_{4 \times 2} \end{bmatrix}, \quad D = \begin{bmatrix} 0_{2 \times 4} \\ I_4 \end{bmatrix}, \quad E = I_2.$$

The prescribed mandatory feedback set is chosen as $\Omega_M = \{(1, 1), (2, 2)\}$, which enforces the corresponding feedback links to remain active throughout the optimization process. These mandatory links represent indispensable local feedback channels that must be preserved, while sparsity is promoted over the remaining entries of the feedback gain matrix.

Sparsity-performance trade-off: To examine the effect of the sparsity-promoting regularization, we solve Problem (P2-C) for a range of values of the regularization weight λ . In particular, λ is swept over the set $\{0\} \cup \text{logspace}(-1.7, 0.1, 20)$,

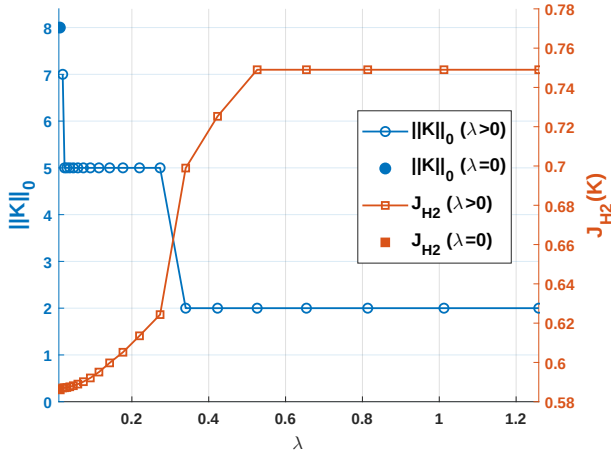


Fig. 1. Buffer network: sparsity–performance trade-off between $\|K\|_0$ and the closed-loop H_2 cost $J_{H_2}(K)$ as a function of the regularization weight λ .

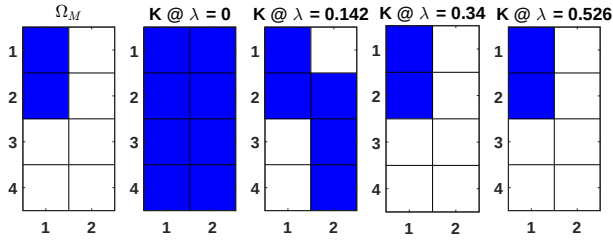


Fig. 2. Buffer network: evolution of the feedback sparsity pattern under the mandatory structure constraint Ω_M .

which corresponds to $\lambda = 0$ together with 20 logarithmically spaced points between approximately 0.02 and 1.26. For each value of λ , the proposed stability-preserving proximal gradient method is initialized using a warm start from the solution obtained at the previous value of λ .

Figure 1 reports the number of nonzero entries of the feedback gain matrix, denoted by $\|K\|_0$, together with the corresponding closed-loop H_2 cost $J_{H_2}(K)$, as functions of the regularization parameter λ . As expected, when $\lambda = 0$, the obtained controller corresponds to a performance-oriented solution under mandatory structural constraints, and therefore achieves the best closed-loop performance but generally yields a denser feedback gain. As λ increases, the feedback matrix becomes progressively sparser, while the H_2 performance degrades gradually. This behavior clearly illustrates the sparsity–performance trade-off captured by the proposed formulation.

Evolution of the feedback sparsity pattern: To visualize the effect of the mandatory structural constraints, Figure 2 shows the evolution of the sparsity pattern of the feedback gain matrix for several representative values of λ , together with the prescribed structure Ω_M , where blue cells represent mandatory connections and white cells correspond to free variables. It can be observed that the entries indexed by Ω_M remain active for all tested values of λ , while sparsification is applied to the non-mandatory entries. This confirms that the proposed method is capable of preserving indispensable feedback links while eliminating redundant feedback channels.

Overall, the small-scale example demonstrates that the proposed method can effectively balance closed-loop performance

and controller sparsity under partial structural constraints.

B. Large-scale mass-spring system

We next consider a large-scale mass–spring system to evaluate the scalability of the proposed method under partial structural constraints. The system consists of $N = 20$ masses connected by springs and dampers with fixed boundary conditions at both ends. Let $q \in \mathbb{R}^N$ and $v \in \mathbb{R}^N$ denote the displacement and velocity vectors, respectively, and define the state as $x = [q^\top v^\top]^\top \in \mathbb{R}^{2N}$. The system dynamics are given by

$$\dot{x} = \begin{bmatrix} 0 & I \\ -M^{-1}K_q & -M^{-1}C_q \end{bmatrix} x + \begin{bmatrix} 0 \\ M^{-1} \end{bmatrix} u + \begin{bmatrix} 0 \\ M^{-1} \end{bmatrix} w, \quad (20)$$

where $M = \text{diag}(m_i)$ is the mass matrix, and K_q and C_q are the stiffness and damping matrices, respectively. Throughout the experiment, we set $m_i = 1$ for all i , while K_q and C_q are chosen as tridiagonal matrices corresponding to nearest-neighbor spring–damper couplings.

The performance output is defined as

$$z = C_z x + D_z u,$$

with

$$C_z = Q^{1/2}, \quad D_z = R^{1/2},$$

where

$$Q = \text{blkdiag}(10I_N, I_N), \quad R = 0.1I_N.$$

We consider a grouped local feedback structure for the mass-spring system, in which every two neighboring masses are treated as one local subsystem. Under this architecture, the mandatory set Ω_M has a hybrid block-diagonal and diagonal form, as illustrated in the top-left panel of Fig. 4. In particular, after partitioning the columns into two parts, the first part is composed of several dense 2×2 diagonal blocks, meaning that the control inputs associated with each subsystem are allowed to use the displacement information of both masses within the same group. The second part only retains diagonal entries, indicating that the velocity feedback is purely local and only depends on each mass itself.

Sparsity–performance trade-off: The regularization parameter λ is swept over the set $\{0\} \cup \text{logspace}(-2, 0.7, 50)$, which corresponds to $\lambda = 0$ together with 50 logarithmically spaced points between 0.01 and 5.01.

Figure 3 reports the number of nonzero entries of the feedback gain matrix, denoted by $\|K\|_0$, together with the corresponding closed-loop H_2 cost $J_{H_2}(K)$, as functions of the regularization parameter λ . When $\lambda = 0$, the unconstrained H_2 -optimal controller is recovered, yielding a dense feedback gain with $\|K\|_0 = 800$ and a closed-loop cost $J_{H_2} = 4.4120$. As λ increases to $\lambda = 0.19$, the feedback matrix becomes substantially sparser, with $\|K\|_0$ reduced to 182, while the closed-loop cost increases only mildly to $J_{H_2} = 4.7541$. This demonstrates that a significant reduction in controller complexity can be achieved with only a marginal loss in performance. However, further increasing the regularization weight leads to a markedly different behavior. For instance,

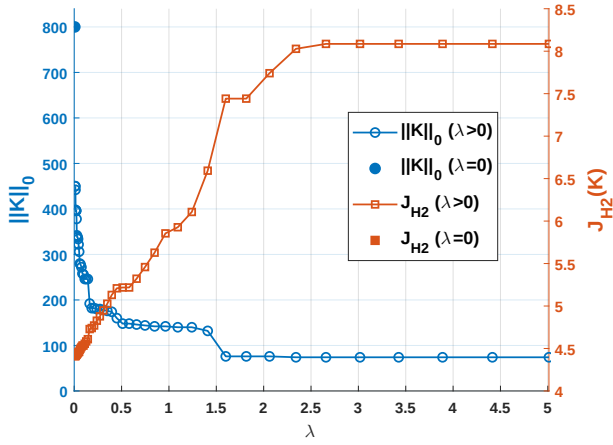


Fig. 3. Mass-spring system: sparsity–performance trade-off between $\|K\|_0$ and the closed-loop H_2 cost $J_{H_2}(K)$ as a function of the regularization weight λ .

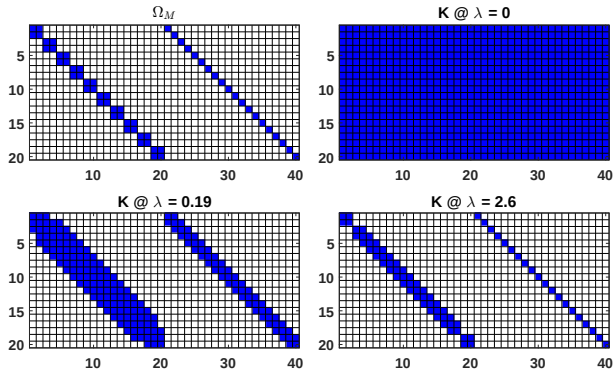


Fig. 4. Mass-spring system: evolution of the feedback sparsity pattern under the mandatory structure constraint Ω_M .

at $\lambda = 2.6$, the number of nonzero entries decreases to $\|K\|_0 = 74$, but the closed-loop H_2 cost increases sharply to $J_{H_2} = 8.3228$, which is nearly twice the cost of the dense baseline. This indicates that excessive sparsification can severely degrade the closed-loop performance, as too many feedback channels are removed. These results again demonstrate the effectiveness of the proposed formulation in capturing the sparsity–performance trade-off.

Evolution of the feedback sparsity pattern: Figure 3 shows the sparsity pattern of the feedback gain matrix for several representative values of λ . It can be observed that the feedback entries indexed by Ω_M remain active throughout the optimization process, while additional non-mandatory entries are gradually removed as the regularization strength increases. This confirms that the proposed method preserves the mandatory structural constraints while adapting the remaining entries according to performance considerations.

Pareto analysis and algorithmic comparison: To further characterize the trade-off between closed-loop performance and controller sparsity, we construct a Pareto front between the closed-loop H_2 cost $J_{H_2}(K)$ and the sparsity level $\|K\|_0$, as shown in Fig. 5. Each point on the curve corresponds to a solution obtained for a specific value of the regularization parameter λ . The Pareto front provides a direct visualization of the achievable sparsity–performance trade-off and facilitates

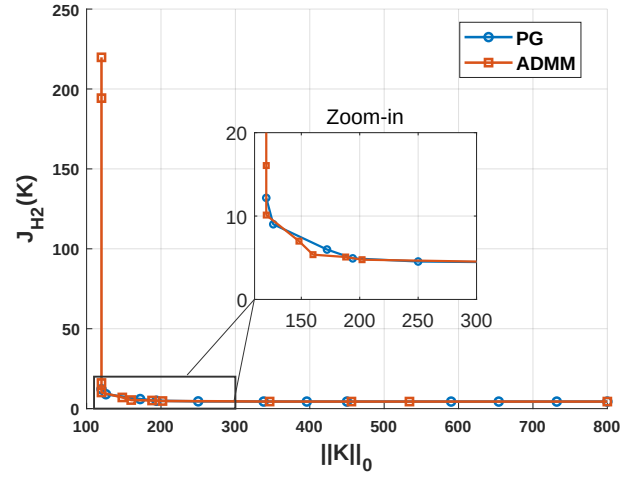


Fig. 5. Pareto front of performance–sparsity trade-off: PG vs ADMM.

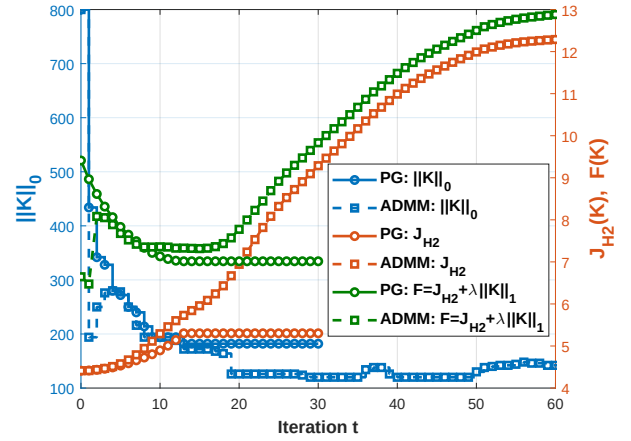


Fig. 6. Convergence comparison between PG and ADMM.

the selection of λ according to practical design requirements.

Figure 5 shows that the relationship between controller sparsity and closed-loop performance is generally highly non-linear. In particular, allowing a small number of additional feedback links may sometimes lead to a significant performance improvement, whereas further densification often yields only marginal gains. This observation indicates that, even after preserving the mandatory feedback links, it remains important to optimize the remaining entries rather than fixing the entire sparse architecture a priori. Therefore, the proposed formulation under partial structural constraints is well motivated from the perspective of sparsity–performance trade-off analysis.

Figure 5 also includes the Pareto front obtained by the ADMM-based method. It is observed that the two methods produce very similar Pareto curves, which indicates that the proposed stability-preserving proximal gradient method achieves solution quality comparable to that of the ADMM-based approach in terms of the sparsity–performance trade-off. The differences between the two methods in the resulting H_2 cost and sparsity level are generally small, suggesting that both methods are capable of identifying competitive sparse feedback designs under the prescribed structural constraints.

Convergence behavior and computational efficiency: Although the two methods yield similar Pareto fronts, their

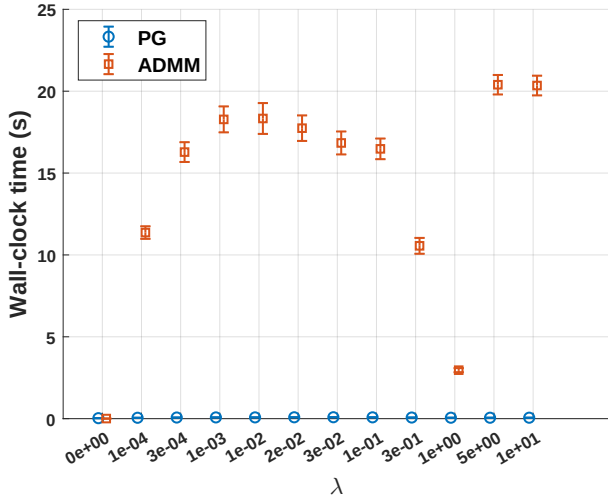


Fig. 7. Computational efficiency comparison between PG and ADMM in terms of wall-clock time.

algorithmic behavior differs significantly. Figure 6 compares the convergence behavior of the sparsity level $\|K\|_0$, the closed-loop H_2 cost $J_{H_2}(K)$, and the composite objective $F(K)$ for a representative regularization weight $\lambda = 0.5$. The proposed method exhibits a more stable and nearly monotone decrease in both the composite objective value and the sparsity level, and reaches a steady regime within a relatively small number of iterations. In contrast, the ADMM-based method shows more pronounced transient oscillations before stabilization and requires substantially more iterations to approach a steady state. This behavior may be attributed to the fact that the proposed method updates the controller gain directly through a proximal gradient step on the composite objective, and the backtracking procedure helps maintain stability and suppress overly aggressive updates. By contrast, the ADMM-based method involves alternating primal updates together with penalty-based coordination, which can lead to stronger transient oscillations in nonconvex and nonsmooth optimization problems.

Figure 7 further reports the wall-clock time of the two methods over different values of λ . It is worth noting that when $\lambda = 0$, the wall-clock times of PG and ADMM are comparable, since in this case the problem reduces to a performance-oriented design without explicit sparsity promotion, and the computational difference between the two methods is less pronounced. As λ increases, however, the efficiency advantage of the proposed PG method becomes increasingly evident. In particular, the PG method consistently requires significantly less computational time than the ADMM-based alternative across almost all tested values of λ . It is also observed that the runtime of ADMM fluctuates considerably with respect to λ , which may be attributed to the sensitivity of its alternating-update scheme and penalty-based coordination to the sparsity pattern induced by different regularization levels. By contrast, the proposed PG method follows a more direct update scheme and therefore exhibits more stable computational behavior. These results further demonstrate the practical advantage of the proposed method in terms of computational efficiency,

especially for large-scale systems.

V. CONCLUSION

This paper studied sparse H_2 state-feedback design under partial structural constraints, where certain prescribed feedback links are required to remain active while sparsity is promoted over the remaining entries of the feedback gain matrix. To handle this problem, we introduced a numerically tractable lower-bound reformulation of the mandatory nonzero constraints and developed a stability-preserving proximal gradient method for the resulting nonconvex and nonsmooth optimization problem. By incorporating a backtracking strategy, the proposed method maintains closed-loop stability throughout the iteration process.

We established the structural feasibility of the mandatory nonzero constraints and showed that, for sufficiently small lower-bound parameters, the reformulated problem admits stabilizing feasible solutions. We further proved that every accumulation point of the sequence generated by the proposed method is a stationary point of the reformulated problem. Numerical experiments on both a small-scale buffer network and a large-scale mass-spring system demonstrated that the proposed method can effectively capture the sparsity-performance trade-off while strictly preserving mandatory feedback links. In comparison with an ADMM-based approach, the proposed method achieved comparable solution quality, but exhibited more stable convergence behavior and significantly improved computational efficiency, especially for large-scale systems.

APPENDIX I DERIVATION OF THE PROXIMAL MAPPING IN SECTION III-C

In this appendix, we derive the entrywise closed-form expression of the proximal mapping associated with the nonsmooth term used in Section III-C.

Recall from Section III-C that the nonsmooth term is defined as

$$g(K) := \lambda \|K\|_1 + I_{\mathcal{C}}(K), \quad (21)$$

where

$$\mathcal{C} := \{K \in \mathbb{R}^{m \times n} : |K_{ij}| \geq \kappa, \forall (i, j) \in \Omega_M\}.$$

Let $\eta > 0$ be a given stepsize, and let $V \in \mathbb{R}^{m \times n}$ denote the input to the proximal mapping. The proximal operator associated with ηg is defined by

$$\text{prox}_{\eta g}(V) := \arg \min_{K \in \mathbb{R}^{m \times n}} \left\{ \frac{1}{2\eta} \|K - V\|_F^2 + g(K) \right\}. \quad (22)$$

Since the squared Frobenius norm and the entrywise ℓ_1 -norm are separable across the entries of K , and the indicator function $I_{\mathcal{C}}(K)$ imposes entrywise constraints on the mandatory entries, the optimization problem in (22) decouples entrywise. Specifically,

$$\frac{1}{2\eta} \|K - V\|_F^2 + g(K) = \sum_{i=1}^m \sum_{j=1}^n \psi_{ij}(K_{ij}; V_{ij}), \quad (23)$$

where

$$\psi_{ij}(x; v) := \begin{cases} \frac{1}{2\eta}(x - v)^2 + \lambda|x|, & (i, j) \notin \Omega_M, \\ \frac{1}{2\eta}(x - v)^2 + \lambda|x| + I_{|x| \geq \kappa}(x), & (i, j) \in \Omega_M. \end{cases} \quad (24)$$

Therefore, for each (i, j) , the proximal mapping is obtained by solving

$$K_{ij}^* = \arg \min_{x \in \mathbb{R}} \psi_{ij}(x; V_{ij}). \quad (25)$$

For non-mandatory entries $(i, j) \notin \Omega_M$, (25) reduces to

$$\min_{x \in \mathbb{R}} \frac{1}{2\eta}(x - v)^2 + \lambda|x|, \quad (26)$$

whose unique minimizer is the standard soft-thresholding operator

$$x^* = S_{\eta\lambda}(v) := \text{sign}(v) \max(|v| - \eta\lambda, 0). \quad (27)$$

Hence, for all $(i, j) \notin \Omega_M$,

$$K_{ij}^* = \text{sign}(V_{ij}) \max(|V_{ij}| - \eta\lambda, 0). \quad (28)$$

For mandatory entries $(i, j) \in \Omega_M$, (25) becomes

$$\min_{x \in \mathbb{R}} \frac{1}{2\eta}(x - v)^2 + \lambda|x| \quad \text{s.t.} \quad |x| \geq \kappa. \quad (29)$$

Ignoring the constraint yields the unconstrained minimizer

$$\tilde{x} = S_{\eta\lambda}(v).$$

If $|\tilde{x}| \geq \kappa$, then $\tilde{x}$ is feasible and hence optimal. Otherwise, the optimal solution must lie on the boundary $|x| = \kappa$. Since the objective is symmetric and strictly convex on each half-line, the optimal boundary point has the same sign as v . Therefore, the unique minimizer is

$$x^* = \text{sign}(v) \max(|v| - \eta\lambda, \kappa). \quad (30)$$

Hence, for all $(i, j) \in \Omega_M$,

$$K_{ij}^* = \text{sign}(V_{ij}) \max(|V_{ij}| - \eta\lambda, \kappa). \quad (31)$$

Combining (28) and (31), we obtain

$$(\text{prox}_{\eta g}(V))_{ij} = \begin{cases} \text{sign}(V_{ij}) \max(|V_{ij}| - \eta\lambda, 0), & (i, j) \notin \Omega_M, \\ \text{sign}(V_{ij}) \max(|V_{ij}| - \eta\lambda, \kappa), & (i, j) \in \Omega_M. \end{cases} \quad (32)$$

This establishes the entrywise update rule in (14).

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